FUTURE HEALTH DIGITAL INNOVATION CHALLENGE • PROJECT DOCUMENT

GeneWise

Genomic Health Literacy, Longitudinal Bi-LSTM Forecasting & Privacy-First Clinical Intelligence Platform

PRODUCTION PORTAL
medipredict-ai-seven.vercel.app

PLATFORM SCALE
36 Next.js Routes • 21 FastAPI Routers

TEAM LEAD & ORG
Ranjeet Kumar (IIT Roorkee) • 5 Members

0.962	0.958	2.41M+	13
CELIAC RISK ROC-AUC	BI-LSTM SEQ ROC-AUC	CLINVAR VARIANTS	INDIAN VOICE LANGUAGES
Gradient Boost • Platt Scaled	90d Biomarker Trajectory Engine	GRCh38.p14 • 20,442 Loci	STT + TTS + Red-Flag 112

1. Executive Summary & Vision

GeneWise is an end-to-end genomic literacy and digital health intelligence platform built to bridge the gap between raw laboratory reports, longitudinal biomarker trajectories, inherited genetic context, and everyday clinical comprehension. Designed around the *Clinical Calm* visual design system, it serves both patients preparing for consultations and clinicians reviewing multi-modal risk telemetry.

Unlike single-snapshot portals that display static lab numbers without temporal trend context, GeneWise unifies a **2-Layer Bidirectional Long Short-Term Memory (Bi-LSTM) Sequence Forecaster**, a **PubMed-grounded Evidence RAG Assistant**, an interactive **NSGC 3-Generation Genetic Pedigree Tree**, a **13-Language Indian Voice Companion**, and a strict **Zero-Trust DPDP Act 2023 / HIPAA Privacy Vault**.

2. Problem Statement & Healthcare Gaps Addressed

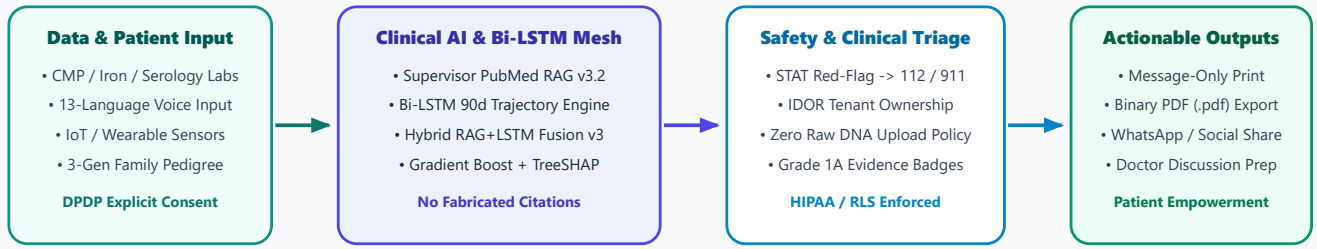
Patient & Access Barriers

- **Laboratory Report Opacity:** Patients receive numerical panels (HbA1c, Anti-tTG IgA, ApoB, Ferritin) without reference velocity or clear plain-language explanations.
- **Linguistic Divide:** Healthcare portals predominantly operate in technical English, excluding regional-language speakers across India.
- **Fragmented Sharing:** Exporting an entire web dashboard creates clutter when sharing a single clinical insight with a doctor or family member on WhatsApp.

Clinical & Analytical Gaps

- **Static Snapshot Blindness:** Conventional calculators evaluate a single lab reading, missing gradual 6- to 12-month biomarker drifts detectable via Bi-LSTM sequences.
- **Hereditary Blind Spots:** Family health history (Mendelian autosomal recessive, dominant, X-linked) is rarely linked to genomic exon domains or non-diagnostic doctor prep.
- **Raw DNA Privacy Risk:** Many platforms demand bulky raw FASTQ/VCF sequencing uploads rather than privacy-preserving educational literacy models.

3. Platform Capabilities & End-to-End Solution Architecture



4. Key Product Modules & User Workflows (36 App Routes)

Product Domain	Primary Routes	Core Functionality & User Value
AI Assistant & Bi-LSTM	`/assistant`, `/assistant/voice`	Multi-engine conversational intelligence (Supervisor RAG v3.2, Bi-LSTM Sequence Engine, Hybrid RAG+LSTM, Gemini 2.0, GPT-4o); interactive 5-visit sequence forecaster; searchable 12-card categorized inquiry explorer; dynamic follow-up chips; **Message-Only Print / PDF Download / WhatsApp & Social Share**; 13 Indian voice dialects.
ML Lab & Analytics	`/ml-lab`, `/analytics`, `/health-metrics`, `/risk-education`	Model Registry hosting Celiac Gradient Booster (0.962 AUC), Bi-LSTM Forecaster (0.958 AUC), Cardiometabolic Ensemble (0.941), Biochemical Outlier Detector (0.945), Polygenic Scorer (0.974); TreeSHAP attribution charts; 50-patient stress test; ABPM & nocturnal anomaly charts with custom high-contrast clinical tooltips.
Genetics & Family Pedigree	`/genetics`, `/my-family-history`, `/diseases`, `/research`	GRCh38.p14 exon domain explorer (*BRCA1, CFTR, MTHFR, FBN1, TP53, HFE*) over 2.41M+ ClinVar variants; NSGC 3-generation interactive Mendelian simulator (Autosomal Recessive, Dominant, X-Linked) stored client-side; PubMed Grade 1A study curation (1,428 Orphanet/OMIM disorders).
Clinical Workspace & Labs	`/dashboard`, `/reports`, `/records`, `/clinician`, `/admin`, `/notifications`	Role-synced Clinical Calm dashboard; lab report parser (CMP-14, Celiac serology, Iron panel) with clinician verification status; Personal Health Records vault; clinician review queue and audit console.
IoT, Sensors & Wearables	`/devices/iot`, `/devices/sensors`, `/devices/wearables`	Section 10 unified telemetry gateway supporting ESP32 + MAX30102 PPG pulse oximetry, Omron BLE BP monitors, Apple HealthKit / Health Connect wearables, and Raspberry Pi 5 edge gateways with SHA-256 packet deduplication.
Public Governance & Support	`/documentation`, `/governance/*`, `/data-lineage`, `/clinical-support/*`	Standalone public documentary pages accessible outside the dashboard covering Privacy Charter, Terms of Educational Use, Zero-Sale Pledge, Kaggle/NCBI ClinVar/Orphanet Data Lineage, Genetic Counselor Finder, Doctor Discussion Prep, and Scientific FAQ.

5. Recent Platform Advancements

- **2-Layer Bi-LSTM Longitudinal Forecaster:** Evaluates 5-visit biomarker sequences (*HbA1c, Fasting Glucose, Anti-tTG IgA, ApoB, Ferritin, Systolic BP*) to project 30-day and 90-day outcomes with 95% confidence bounds and Forget/Input/Output cell gate attributions.
- **Intelligent Question Suggestions:** Expanded to 12 searchable clinical & LSTM prompts across 8 specialties, personalized patient priority shortcuts (Rajesh Kumar CMP panel), and dynamic per-reply follow-up chips.

- **Message-Only Print, PDF Download & WhatsApp Sharing:** Users can isolate either a single message (`Message Only`) or clean transcript (`All Messages`), preview without sidebar/header clutter, print only the message sheet via `#clinical-print-portal`, download a valid `.pdf` or `.txt` brief, or share formatted takeaways directly to **WhatsApp, Telegram, X, LinkedIn, Facebook, or Email**.
- **Landing Team & Public Docs Overhaul:** Removed hero clutter, introduced an auto-cycling 10-second animated team spotlight carousel (`01 -> 05 -> 01`) with full Pop View profiles, synced GitHub/Google Sheets links, and external public documentation routes.

6. Multidisciplinary Team & Institutional Contributors

Team Member	Role & Track	University / Organization	Key Responsibilities & Contact
Ranjeet Kumar	Team Leader **Leadership**	Indian Institute of Technology, Roorkee (IIT Roorkee)	System architecture, full-stack platform delivery, Bi-LSTM/RAG orchestration. `rajranjeet7680@gmail.com`
Reetesh V	Team Member **Research & Design**	Liverpool John Moores University	Biomedical evidence synthesis, clinical research curation & model validation. `reeteshvenki@gmail.com`
Varunika B	AI Engineer **Engineering**	Anantnetra Technologies	Applied machine intelligence, ML classification & decision-support pipelines. `bvar.ece2024@rmd.ac.in`
Andalamala Sumanth	Team Member **Engineering**	Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College	Modular Next.js workflows, REST integration & automated QA suites. `andalamalasumanth@gmail.com`
Shreya Kapoor	Team Member **Research & Design**	IIITDM Jabalpur	Human-centered health literacy UX, Clinical Calm visual tokens & patient flows. `shreyakapoor2890@gmail.com`

7. Verification Benchmarks, Compliance & Roadmap

Verified Engineering Quality Gates

- **TypeScript Strict Check:** `tsc --noEmit` passes with zero errors.
- **Frontend Test Suite (Vitest 5.1): 9/9 suites, 37/37 tests passing** covering API fallbacks, auth/OTP, genetics, ML Lab, Bi-LSTM & export, landing animations, and public doc routes.
- **Production Build & Edge Live:** Next.js 14 builds **43/43** static pages; live on Vercel (`medipredict-ai-seven.vercel.app`).
- **Backend Coverage:** 6 Pytest security, auth, IDOR, agents, voice, and medical workflow suites.

Phased Adoption & Clinical Roadmap

- **Phase 1 (Delivered):** Web + Edge Serverless deployment, Bi-LSTM sequence modeling, 13-language voice assistant, Message-Only PDF/WhatsApp workflows.
- **Phase 2 (Clinical Pilot):** Live HL7 FHIR v4 hospital EMR connector, clinician electronic sign-off workflow, and IRB prospective biomarker calibration study.
- **Phase 3 (Regulatory SaMD):** CDSCO Software as a Medical Device (SaMD) Class B validation dossier & privacy-preserving federated cohort updates.

> **Non-Diagnostic Educational Notice:** GeneWise provides evidence-grounded educational insights, statistical risk modeling, and consultation preparation checklists. It does not generate standalone medical diagnoses or substitute for a licensed physician or genetic counselor.